

Smooth compact perturbations preserve the determinant integral difference

MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff

theorem

Builds a finite telescope from single-output perturbations with compactly supported fluxes.

Statement

Fix $n \geq 1$ and let $g, u : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^∞ , with u compactly supported. Then

$$\int_{\mathbb{R}^n} (\det D(g + u)(x) - \det Dg(x)) dx = 0.$$

Assumptions

Both maps are smooth of all finite orders, as stated in the exact theorem. Only u must have compact support. No decay or global integrability is assumed for g or for its determinant separately. Here and below $g_j(x) := (g(x))_j$ and $u_j(x) := (u(x))_j$ are scalar coordinate maps. Compact support means that $\text{supp } u := \overline{\{x : u(x) \neq 0\}}$ is compact.

Conclusion

The determinant difference is integrable and has integral zero. This statement is an integral of a difference; it does not assert equality of two separately integrable whole-space determinants.

The one-coordinate integral lemma needs only a C^2 base map and a C^1 compactly supported scalar perturbation. The theorem here retains its stronger stated C^∞ hypotheses and applies that lemma to each smooth hybrid.

Proof route

Add the components of u one at a time, prove each determinant increment integrable with zero integral, and add the finitely many increments.

▼ Proof steps

1. For $S \subseteq \{1, \dots, n\}$, define the smooth hybrid map by its coordinates:

$$(g_S(x))_j = \begin{cases} g_j(x) + u_j(x), & j \in S, \\ g_j(x), & j \notin S. \end{cases}$$

Then $g_\emptyset = g$, $g_{\{1, \dots, n\}} = g + u$, and for $i \notin S$,

$$g_{S \cup \{i\}}(x) = g_S(x) + u_i(x)e_i.$$

Each scalar coordinate u_i is smooth and has compact support, since

$$\{x : u_i(x) \neq 0\} \subseteq \{x : u(x) \neq 0\} \implies \text{supp } u_i \subseteq \text{supp } u.$$

2. Apply the component-flux identity with base g_S , scalar u_i , and row i . Smoothness supplies the required C^2 and C^1 hypotheses. The flux $u_i C_i(Dg_S)$ is C^1 and compactly supported because its support is contained in that of u_i . The compact-support divergence theorem gives

$$\int_{\mathbb{R}^n} (\det Dg_{S \cup \{i\}} - \det Dg_S) = 0.$$

This increment is integrable: it is continuous and vanishes outside the compact support of u_i , since there u_i is locally zero and its derivative is zero.

3. Choose any ordering of the n coordinates and write its successive subsets as $S_0, \dots, S_n$. Pointwise telescoping gives

$$\det D(g + u) - \det Dg = \sum_{r=0}^{n-1} (\det Dg_{S_{r+1}} - \det Dg_{S_r}).$$

Every summand is integrable by the preceding step. Linearity of the integral is therefore legitimate for this finite sum, and its integral is the sum of n zeros. No splitting into the two unrestricted determinant integrals is used.

Main citations

- Exact declaration and its source — `MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff`
- Pointwise component-flux identity — `MathlibAnnex.Piola.divergence_componentFlux_eq_det_sub`
- Integral of a compactly supported divergence — `MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport`
- Hybrid with a finite subset of perturbed outputs — `MathlibAnnex.Piola.outputHybrid`
- Empty hybrid is the base map — `MathlibAnnex.Piola.outputHybrid_empty`
- Full hybrid is the perturbed map — `MathlibAnnex.Piola.outputHybrid_univ`
- Zero integral for a single-output increment — `MathlibAnnex.Piola.integral_det_singleOutputPerturb_sub_eq_zero`
- Adding one coordinate to a hybrid — `MathlibAnnex.Piola.outputHybrid_insert_eq_singleOutputPerturb`

Lean source signature (exact)

```
theorem integral_det_fderiv_add_sub_eq_zero_of_contDiff
  {m : ℕ} {g u : (Fin (m + 1) → ℝ) → (Fin (m + 1) → ℝ)}
  (hg : ContDiff ℝ (↑(T : ℕ∞)) g) (hu : ContDiff ℝ (↑(T : ℕ∞)) u)
  (huc : HasCompactSupport u) :
```

$$\int x, (\text{LinearMap.det } ((\text{fderiv } \mathbb{R} \text{ (fun } y \Rightarrow g \ y + u \ y) \ x)).\text{toLinearMap}) - \text{LinearMap.det } ((\text{fderiv } \mathbb{R} \ g \ x).\text{toLinearMap})) = 0$$

In the source	Mathematical meaning
<code>{m : ℕ} {g u : (Fin (m + 1) → ℝ) → (Fin (m + 1) → ℝ)}</code>	The positive dimension $n = m + 1$, base map g and perturbation u on $\mathbb{R}^n$.
<code>(hg : ContDiff ℝ (↑(T : ℕ)) g) (hu : ContDiff ℝ (↑(T : ℕ)) u)</code>	Both maps are C^∞ , meaning continuously differentiable at every finite order.
<code>(huc : HasCompactSupport u)</code>	Only u is required to have compact support $\overline{\{x : u(x) \neq 0\}}$.
<code>fun y => g y + u y</code>	The pointwise sum $g + u$, whose derivative is used in the first determinant.

In the source	Mathematical meaning
<code>LinearMap.det ((fderiv ℝ (fun y => g y + u y) x).toLinearMap) - LinearMap.det ((fderiv ℝ g x).toLinearMap)</code>	The entire integrand $\det D(g + u)(x) - \det Dg(x)$; these are signed determinants of the same-size derivative maps.
<code>∫ x, (LinearMap.det ((fderiv ℝ (fun y => g y + u y) x).toLinearMap) - LinearMap.det ((fderiv ℝ g x).toLinearMap)) = 0</code>	The conclusion integrates this difference over all $\mathbb{R}^n$ with Lebesgue measure and gives zero. It does not assume that the two separate whole-space determinant integrals are finite.

▼ Exact surrounding binder context (separate excerpt)

noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators Topology

namespace MathlibAnnex
namespace Piola

▼ Exact content identity

Declaration: MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff

Accepted content SHA-256: a793404962f66d63ae5b1099565e0fcb6b449d11285c7f82cf00c7117b2ef924

Accepted source guide SHA-256: 3b922fceda2d66468f42d46e654bb7524c00fdb0a0f47c4a94db8fad7d5e21f1

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73